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Drone-Based Quantum Key Distribution (QKD)

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) are used in numerous applications ranging from defense, law enforcement, environmental monitoring, disaster recovery, aerial photography, and delivering consumer packages. Securing wireless communication between drones in-flight is critical to ensure safe operation during flight and avoid multiple types of attacks, *e.g.*, eavesdropping, spoofing, jamming, *etc*. Quantum communication protocols offer enhancements over classical approaches. In this effort, we present progress towards demonstrating Quantum Key Distribution (QKD) between two drones in flight. A significant challenge includes achieving system performance using compact Size, Weight, and Power (SWaP) constraints of the drone vehicle. We introduce and evaluate critical subsystems including the QKD source, which is based on resonant-cavity Light Emitting Diodes (LED) controlled by an FPGA, and we discuss a secondary QKD source based on a fiber-coupled polarization modulator. The Pointing, Acquisition and Tracking (PAT) system is comprised of several cascading subsystems, which provide course alignment using based on Infrared (IR) beacons/cameras with gimbals, and fine alignment using Fast Steering Mirrors (FSM) with absolute encoders and feedback position sensors. We discuss both transmit and receive optics including custom designed 3D-printed optical benches. Finally, we introduce single-photon detectors, FPGA-based time-tagger, and a novel statistical post-processing synchronization algorithm. Establishing a quantum communications link between drones in-flight is an important prerequisite for future drone-based quantum applications such as entanglement distribution, distributed quantum sensing, and Quantum Positional Verification (QPV).

Keywords: Quantum Key Distribution (QKD), Quantum Communications, Multi-rotor Drone, Quantum Cryptography

1. INTRODUCTION

Aerial drones are being leveraged to solve many practical problems in our modern society, including providing near real-time delivery of goods and services (including the Covid-19 vaccine), as well as humanitarian relief groups, municipal governments, law enforcement, and Department of Defense (DoD), *etc*. Drones offer many benefits (*e.g*., nearly direct line of sight, unmanned operation, relatively low cost, *etc*.). However, drones are known to be vulnerable to certain attack vectors (*e.g*., command and control signals) by adversaries who seek to disrupt operations, divert the trajectory of drone, or cause an intentional crash. Securing drone communications channels is of great importance, especially as drones become more ubiquitous and the airspace becomes more congested with drones carrying packages (and possibly carry humans via drone taxis in the future). One approach to securing communications is using quantum approaches such as Quantum Key Distribution (QKD). Establishing quantum communications links between mobile platforms has challenges. However, previous researchers have demonstrated QKD from a plane to ground [1], ground to plane [2], and satellite to ground [3]. Additionally, quantum communication links from a drone to ground have been demonstrated [4-5]. In this paper, we report progress towards drone-to-drone QKD, which has yet to be demonstrated. Such reconfigurable links are also step toward more general entanglement-based quantum networking, where they can link fiber-based networks, couple to other free-space applications, and enable distributed quantum sensing and computing.

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2. PROPOSED SYSTEM ARCHITECTURE

Drone-based QKD requires multiple critical subsystems operating under restrictive Size, Weight, and Power (SWaP) constraints on the drone. Our current system architecture consists of two commercial off-the-shelf (COTS) unmanned aerial systems (UAS): a QKD Transmitter Drone (TX - Alice), and QKD Receiver Drone (RX - Bob). The TX drone consists of a compact and mobile Decoy-State QKD source [6-7], which produces three-state QKD signals and employs decoy states to provide mitigation against photon number splitting attacks. We have developed two compact QKD sources, one based on Resonant Cavity LEDs, and the other based on polarization modulators. The QKD states are collimated and transmitted over the free-space optical path between the TX and RX drone, using custom 3D-printed optics assemblies and a novel Pointing, Acquisition, and Tracking (PAT) system. The RX drone consists of reciprocal PAT system and 3D-printed receive optics, which perform the required quantum state projections; a 4-channel single-photon detector is employed on the RX drone to detect the QKD states in two mutually unbaised bases. Detector clicks are recorded using a custom FPGA-based time tagger. To synchronize the QKD protocol, we are developing two approaches: employing Software Defined Radios (SDR) and using the QKD detections themselves. Here we present the critical subsystems, and characterization results to demonstrate progress towards drone-based QKD.

3. SYSTEM CHARACTERIZATION RESULTS

3.1 Drone Platform

We use COTS multi-rotor drone platforms (Model Number: DJI S1000+), modified to integrate QKD-critical subsystems (see Fig. 1). The S1000+ has a gross takeoff weight of 11 kg, with a maximum payload capacity of 6.8 kg. We designed a power tether system to provide continuous flight operation without the need to land periodically to swap batteries. The optics payload and PAT sensors reside in a 3-axis stabilization gimbal (Model Number: Gremsy T3). Critical subsystems are mounted into the drone platform and draw power from the main power bus of the drone using appropriate fusing, DC/DC conversion, and filtering.

Figure 1. Drone-based QKD Hardware platform (DJI S1000+), and QKD subsystems.

3.2 QKD Source (Resonant Cavity LED's)

The QKD source in the TX drone produces three QKD states, encoded in the polarization degree-of-freedom (DOF): Right-Circular and Left-Circular (for encoding), and Horizontal (for error checking). The circular basis is selected due to its resilience against frame-of-reference rotations between drones while in flight. Three resonant cavity LEDs (RC-LED, Model Number: RC650-TO46FW) are used to produce the QKD states. Because each QKD state is produce by a unique RC-LED, great care is taken to ensure that the produced state is indistinguishable among all other DOF's (*e.g.*, spectrum, time of arrival, *etc*.). Characterization testing is performed to compare the spectrum and temporal response of the RC-

LEDs, which are seen to have high spectral and temporal overlaps; see Fig. 2. Additional spectral indistinguishability can be achieved using a narrow bandpass filter.

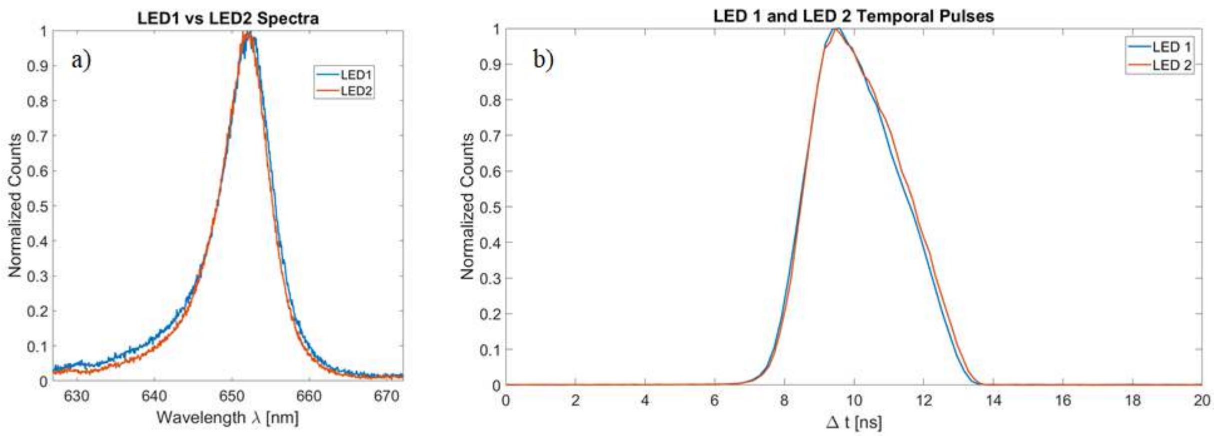


Figure 2. QKD Source Characterization. a.) Spectra and b.) temporal response of two resonant cavity LEDs.

The resonant cavity LEDs are driven using an FPGA through a daughter card, pictured in Fig. 3a). Logic signals from the FPGA are routed down different paths to drive each RC-LED with a higher or lower current to produce signal- and decoy-state levels, respectively. The QKD source uses a QKD clock rate of 25 Mbits/sec. Quantum photonic states generated by the RC-LEDs are coupled into single-mode fiber using a custom 3D-printed mounting fixture (Fig. 3b). Set screws positioned on the 3D-printed fixture enable fine adjustment to maximize coupling light from the RC-LEDs to the single-mode fibers, which are then routed to the TX optical bench for quantum state preparation.

a.)

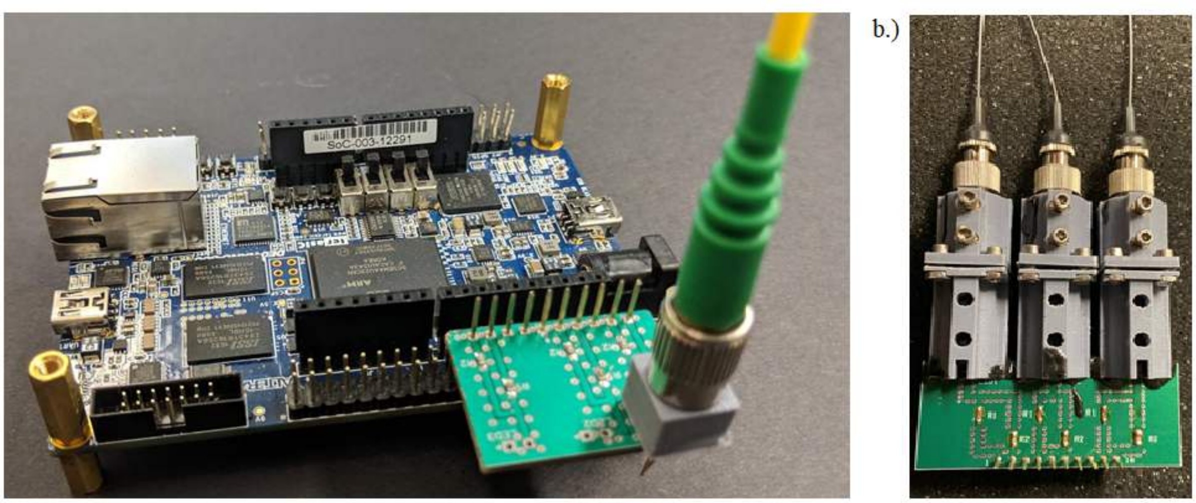


Figure 3. QKD Source Hardware. a.) Prototype-build FPGA, RC-LED driver board, and a single fiber optical channel; b.) final configuration of three RC-LEDs coupled into single-mode fiber.

3.3 QKD Source (Polarization Modulator)

In order to eliminate any possible side-channel attacks that may persist with the RC-LEDs (e.g., if the relative photon emission times happened to drift), we are also pursuing an alternative QKD source, using a custom KTP Polarization Waveguide Modulator produced by AdvR (Bozeman, Montana, USA). The concept of operation consists of launching diagonally polarized light into the KTP waveguide. Applying a voltage pulse to the crystal applies a relative phase to the vertical component of the wavepacket. By suitable selection of voltage amplitude, a relative phase of $+\pi/2$, $-\pi/2$, or 0 is applied to the quantum state, producing Right Circular, Left Circular, or Diagonal polarization, respectively. The KTP has intrinsic birefringence, which has the effect of temporally separating the horizontal and vertical components of the wavepacket; if this separation exceeds the coherence length of the photons (proportional to the inverse bandwidth of the light), the photons emitted will be in a mixed state of polarization. Therefore, we must incorporate a suitable compensation crystal after the KTP, allowing the horizontal and vertical components to be temporally overlapped.

The characterization setup of the polarization modulator-based QKD source is presented in Fig. 4. A HeNe laser launches linearly polarized light into a polarization maintaining fiber (PMF); AdvR aligned the PMF to connect to the modulator at 45° angle with respect to the KTP fast axis. The KPT crystal is driven by an amplified voltage signal, which induced a phase term on the vertical component of the polarization state. For device characterization, an RF signal is used; in actual QKD operation the pulse amplitudes would be applied randomly (to create the three polarization states). A high-speed oscilloscope records the projection measurements made by a Si photodetector after a polarizer and quarter waveplate. The fidelity with the target states (Right-, Left- and Diagonal-polarized) as a function of time is presented in Fig. 5. From our measurements we can infer an equivalent quantum Bit Error Rate below 1.5% for all states.

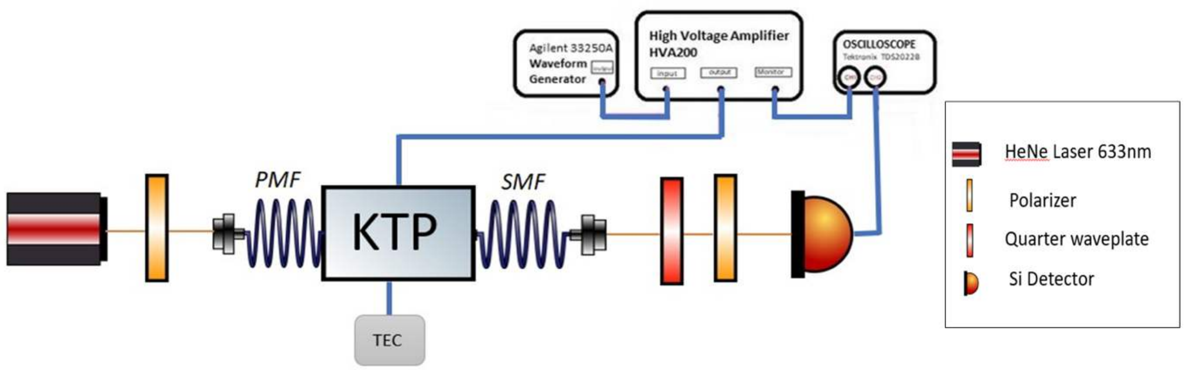


Figure 4. QKD Polarization Modulator source characterization setup. A HeNe laser passes through a polarizer aligned with the stress-rods of a polarization maintaining fiber (PMF) which is fused to the KPT crystal (temperature stabilized using a Thermal Electric Cooler (TEC)) at a key angle of 45°. The output light is projected onto various polarizations and measured with a silicon photodiode. RF drive voltages for the KPT crystal are produced by a waveform generator and an amplifier.

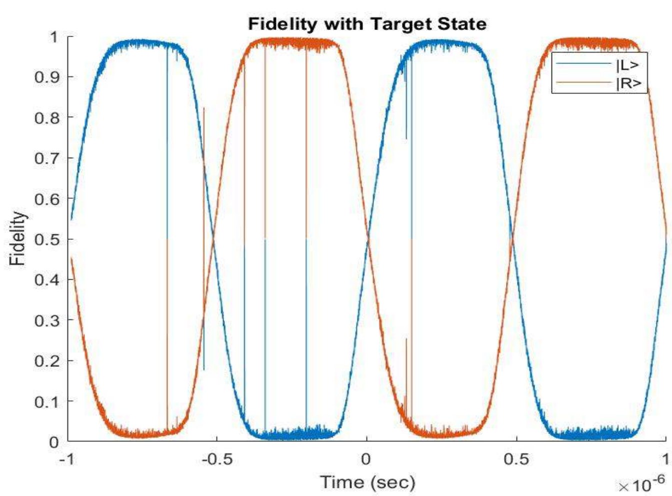
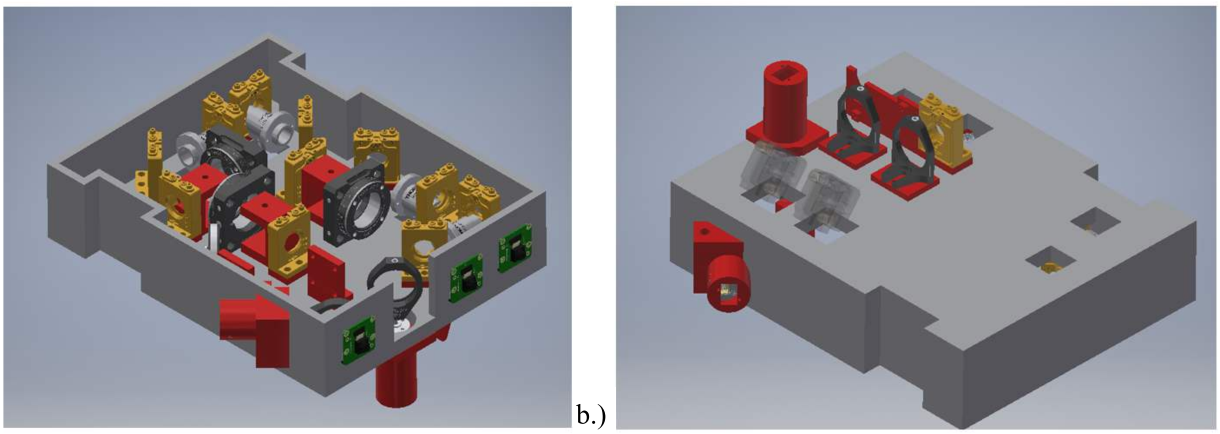


Figure 5. QKD Polarization Modulator target states of Left- and Right-Circular polarization, as a function of time (Characterization Testing).

3.4 TX/RX Optics

Custom 3D-printed optical benches are developed for both quantum state preparation (TX drone) and state projection (RX drone). To make a compact optical bench, the optical components are mounted on both sides of the benches. PAT subsystem components including Fast Steering Mirrors (FSM), beacon lasers, IR Beacon, IR Camera, Long-Range IR Camera, and Position Sensitive Diode (PSD) are also integrated into the TX and RX optical benches. A 3D model of the TX/RX optical bench design for the top and bottom are presented in Fig. 6a), and 6b), respectively. Each optical bench is located on the 3-axis gimbal in the respective drone.



a.) b.)

Figure 6. TX and RX custom 3D-printed optical benches. a.) Top and b.) bottom view of optical bench.

3.5 PAT Subsystem

Quantum states that are produced in the TX drone must be sent successfully to the RX drone for measurement. Mechanical vibrations due to the drone's motors and random air turbulence, in part induced by the drones themselves, complicate establishing and maintaining a stable line-of-sight between drones; thus, active stabilization control is required. The PAT subsystem performs active stabilization in a cascaded control system consisting of an outer control loop, which provides course alignment control, and an inner control loop, which provides fine alignment control. An overview of the PAT subsystem is presented in Fig. 7. The outer control loop uses IR cameras to sense the relative position of the opposite drone, which broadcasts an IR beacon (850 nm). Based on the relative position of the IR beacon in the image, a compensating control signal is used to steer the 3-axis gimbal, which directs the location of the IR beacon to the center of the image. The frame rate of the IR camera is 97 frames per second.

The inner control loop uses bi-directional laser beacons, fast steering mirrors (FSM, Model Number: LR17), and position-sensitive detectors (PSD, Model Number: 5000011). The TX drone transmits a 710-nm laser beacon to a PSD on the RX drone, producing a position error signal that is sent over a Wi-Fi link back to the TX drone. The error signal is used to steer the TX drone's FSMs to center the beacon location on the setpoint on the PSD. Fig. 8 shows an example PSD Calibration test; during the test, the setpoint was stepped in a grid pattern while the control loop attempted to lock the beam's location to the setpoint.

The inner control loop FSMs have integrated absolute magnetic encoders, which enable steering the FSM to an arbitrary pointing vector (±2 mrad). These absolute magnetic encoders are used during initial acquisition and re-acquisition processes, in which a long-range IR camera senses the IR beacon on the opposite drone and steers the FSM to the expected location of the input aperture of the opposite drone. This capability reduces both the initial acquisition time and re-acquisition times to less than 1 sec (assuming the IR beacon of the opposing drone is visible to the long-range IR camera).

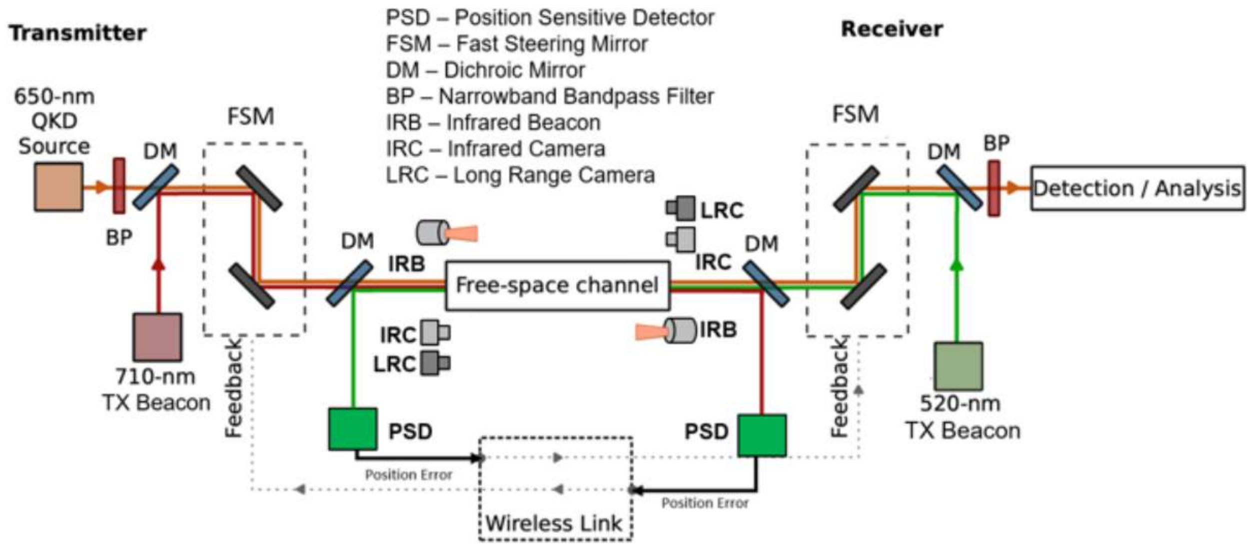


Figure 7. PAT subsystem. Course alignment is achieved using IR beacons and IR cameras providing angular corrections using the Gimbal. Fine alignment is achieved using bi-directional beacons (520 nm and 710 nm) between drones, PSD and FSM's. PSD sensors ascertain the incident beam's position, and error signals are relayed using a WiFi link between drones; these signals provide the control input for the FSMs in the opposite drone.

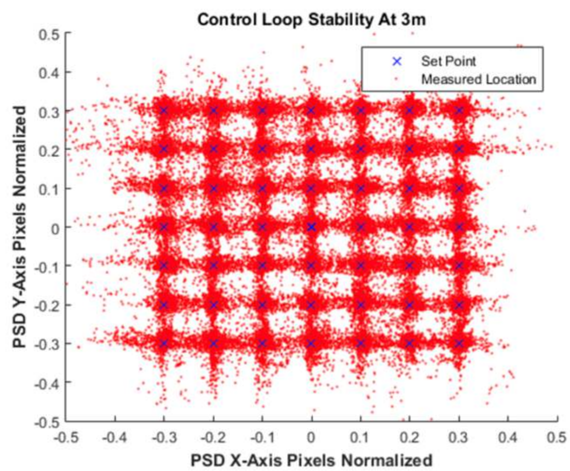


Figure 8. Inner Control Loop PSD Calibration test. Blue X's indicate the set point, while red points are the measured location. Drone separation was 3 meters. Setpoints were swept in a grid pattern over time, and the tracking system followed the setpoints, producing vertical and horizontal streaks during the calibration test.

3.6 Single-Photon Detectors

Once the quantum states are transmitted from the TX drone to the RX drone, polarization projection measurements are performed, measured using single-photon detectors on the RX drone; to minimize SWaP, we use a 4-channel single-photon detector (Excelitas Model Number SPCM-AQ4C). The detectors have ~60% detection efficiency at the quantum channel wavelength and can achieve continuous count rates above 1.5 Mc/s. A separate interface board provides signal conditioning and power inputs for the AQ4C unit; a custom cable assembly enables the interface board to connect in an inverted orientation, to reduce the overall size. Appropriate power supply circuits provide filtered power to the single-photon detectors using the main power bus of the drone.

3.7 FPGA Time Tagger

The single-photon detectors output a logical high voltage pulse (4.5 V) after receiving each detection event. These logical pulses are routed to an FPGA (Intel Model Number: DE-10 Standard) programmed to serve as a time tagger. Detection events on each channel are recorded using a 32-bit time tag and are saved to an on-board SD card for post-processing. The FPGA records detection events using a clock frequency of 100 MHz. Filtered DC power for the FPGA time tagger is provided by the main power bus of the drone.

3.8 Time Synchronization

In QKD, Alice and Bob must be able to compare Bob's measurement basis for a given qubit with the basis in which Alice prepared it. It would be meaningless to compare these for two different qubits, so it becomes necessary for Alice and Bob to reconcile their clocks. Thus, synchronization between the transmitter and receiver is required to perform the appropriate post-processing of the QKD data. Such time synchronization is complicated due to both drones being mobile platforms, which experience relative motion with respect to each other. Our first time-synchronization approach uses Software Defined Radios (SDR, Model Number: ARRadio) on both drones. The TX SDR transmits a continuous wave (CW) timing signal to the RX drone. This timing signal is used to align the reference clocks of the FPGA, which drives the QKD source on the TX drone and the FPGA which performs the time-tagging functionality on the RX drone. Because both the QKD signal and the RF-timing signal propagate through space at the speed of light, the quantum and classical signal propagation delays are equal, and the relative motion between the TX and RX drones is compensated for by our SDR time-synchronization approach. An example of SDR characterization is shown in Fig. 9; an I-Q timing signal centered on 2.4 GHz is generated on the TX SDR and measured by the RX SDR.

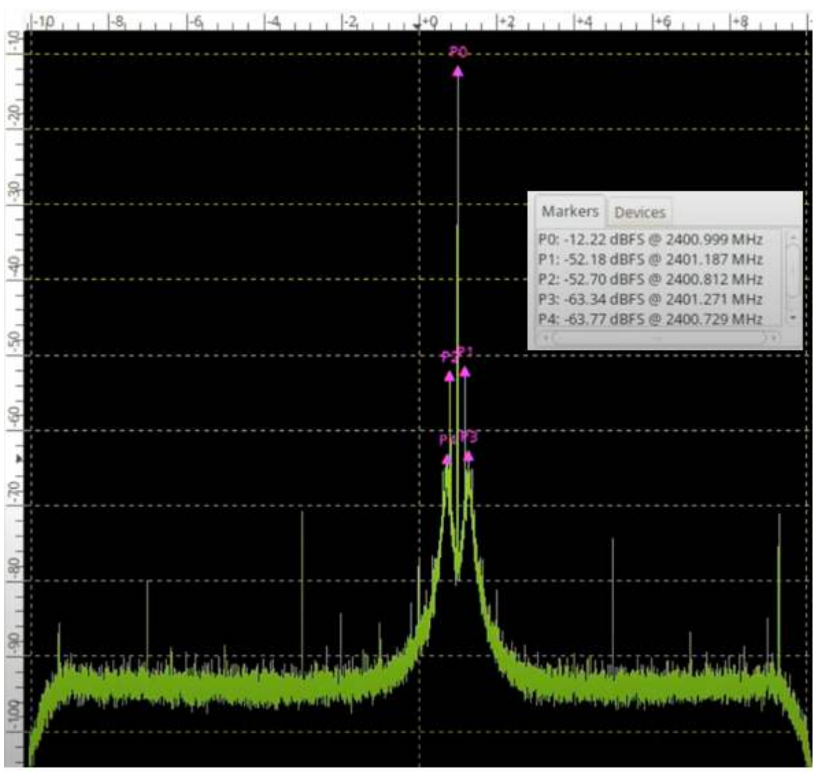


Figure 9. SDR spectrum. 2.4 GHz I-Q Timing Signal. Magnitude and frequency values of the carrier and sideband signals are shown in the inset.

In order to eliminate the need for additional SDR hardware, we have also developed a novel technique for probabilistic qubit-based synchronization during post processing. After the measurement phase is complete, Alice and Bob exchange their basis information over a classical channel in preparation for sifting. At this stage we have Bob compare Alice's published basis choices to his measurement results. Using probabilities derived from known quantities such as the mean photon number, detector dark counts, and the expected polarization of the system, Bob computes the likelihood that a given section of his data has been produced by Alice's published basis choices. By comparing the likelihoods using a Bayesian approach, he can probabilistically determine the correct synchronization. We compute the number of events of each type at potential synchronization points using fast Fourier transforms, and combine this with the probabilities of each event type to output the most likely synchronization point and the uncertainty.

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